

When the depth is set to 0, one object can reflect another, but not much else. Adding more reflections allows objects to reflect one another's reflections, adding to the sense of realism. The number of refractions needed in a scene can add up quickly.

5.4.3 Advanced lighting

Advanced Lighting in the scanline renderer re-creates the way real light scatters through the scene. There are two ways to render advanced lighting effects: by using the Light Tracer or using Radiosity.

The Light Tracer creates soft-edged shadows and simulates colour bleeding for brightly lit scenes. The Light Tracer does not attempt to create a physically accurate model and can be easier to set up than radiosity. The Light Tracer works well with most lights, but adding a skylight will create softer shadows.

5.4.4 Radiosity

Radiosity is more complex than the Light Tracer but is also physically accurate. Radiosity scatters light energy throughout the scene, calculating the light intensity for all surfaces in the environment. It is mostly used for architectural renderings, but can be used for animation. Radiosity relies on the photometric lights in



A scene rendered with radiosity

3ds Max to create its effect. Photometric lights are physically accurate, so they must be placed in a physically correct manner.

Radiosity in 3ds Max works by creating a radiosity solution, which calculates a map of how the lighting affects each and every surface of the scene. If the objects in the scene remain still, the solution can be used for any camera angle. This makes it very useful for an architectural fly through. If any object in the scene moves, however, then the radiosity solution must be recalculated. This can slow rendering significantly when radiosity is used for animation.

5.5 Environment and Effects Panel

The Environment and Effects panel allows you to control such things as the